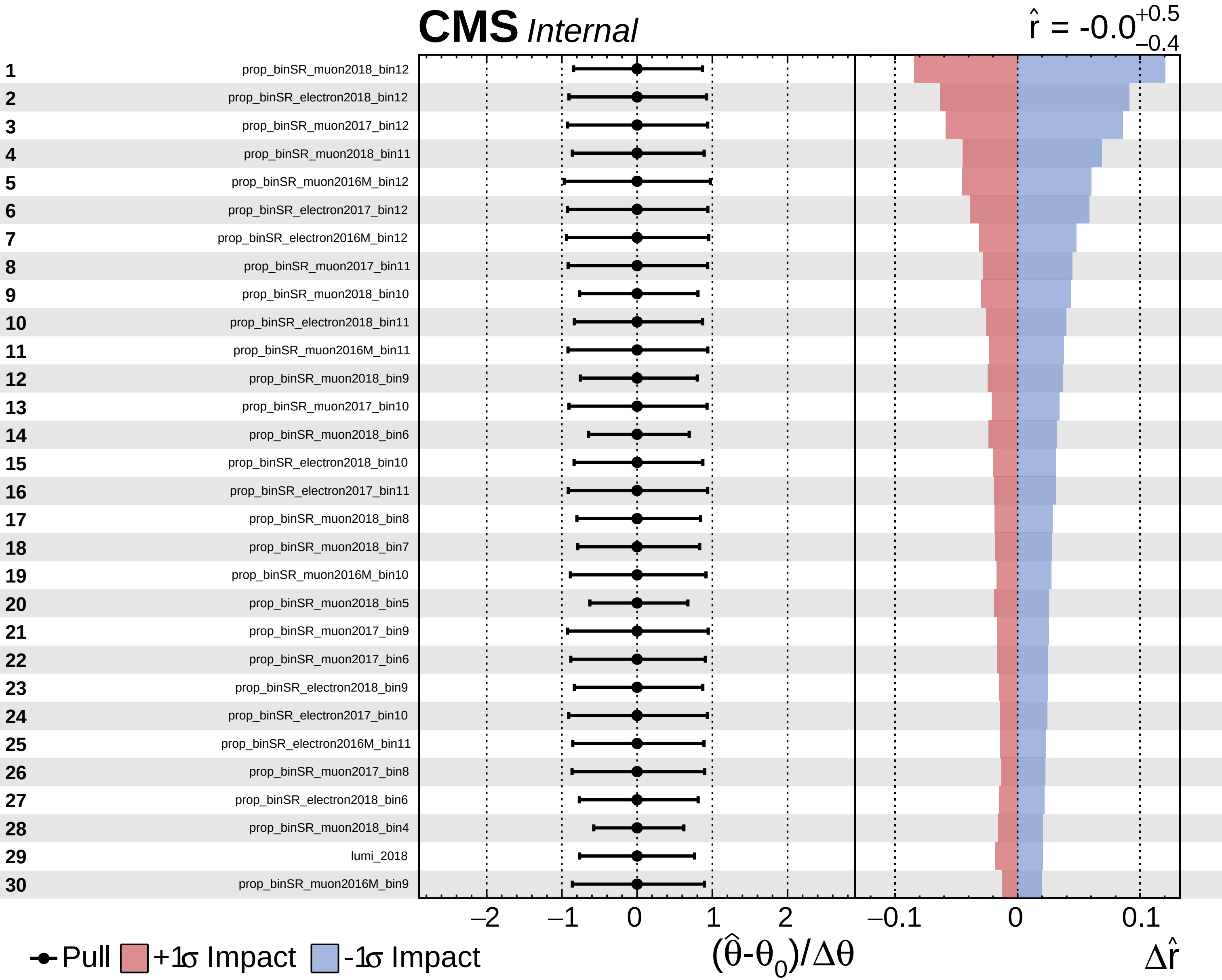
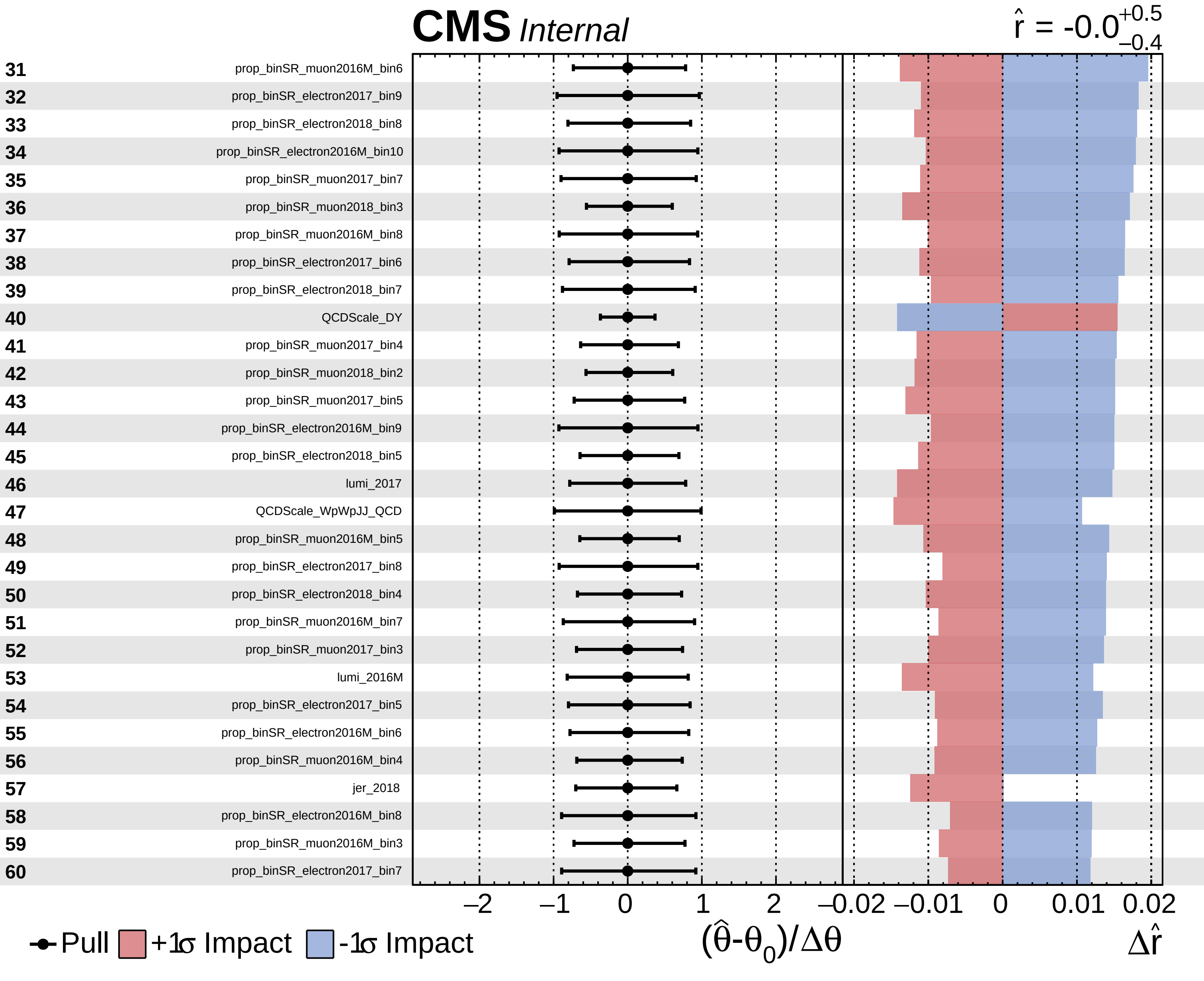
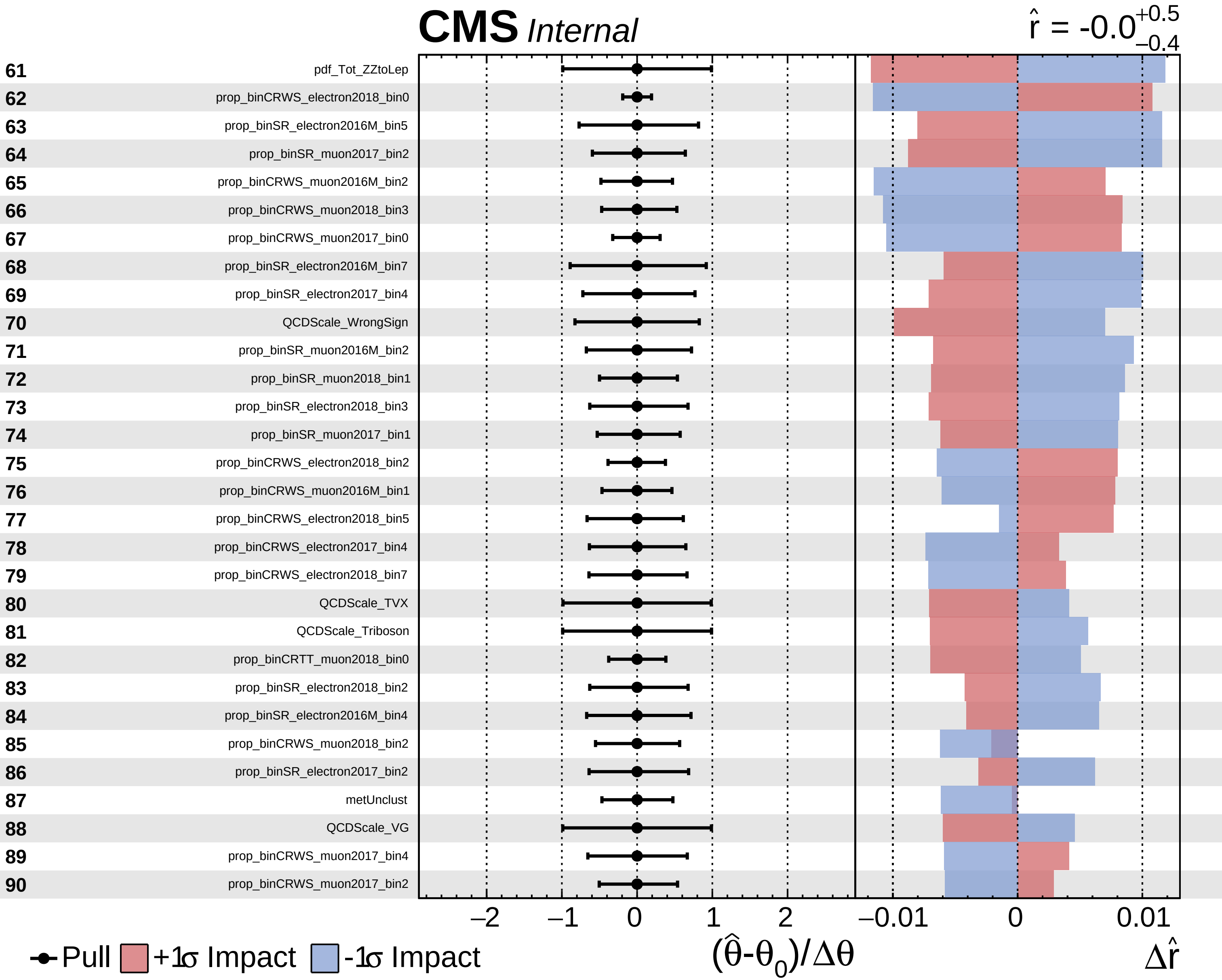
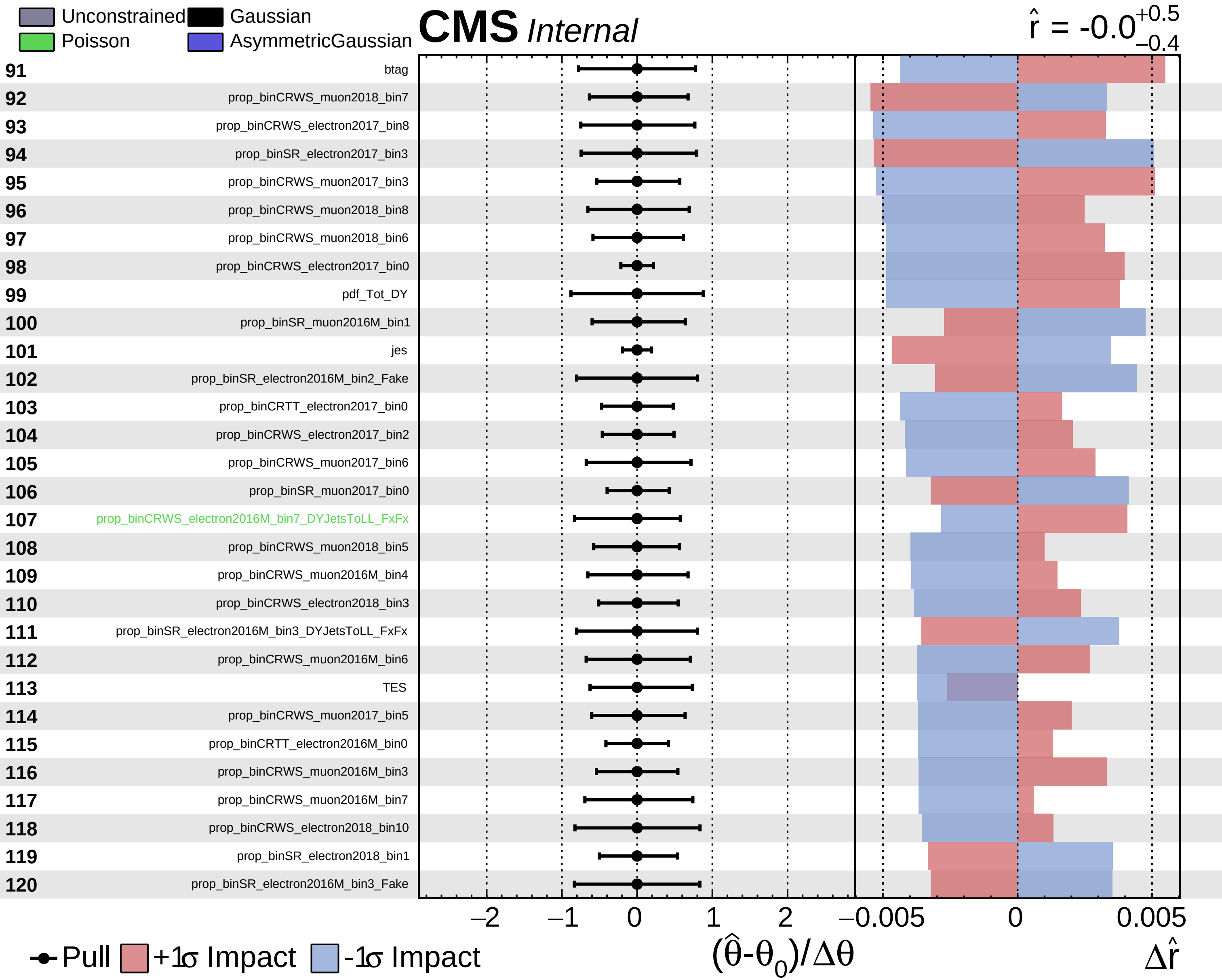


CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
1 prop_binSR_muon2018_bin12
2 prop_binSR_electron2018_bin12
3 prop_binSR_muon2017_bin12
4 prop_binSR_muon2018_bin11
5 prop_binSR_muon2016M_bin12
6 prop_binSR_electron2017_bin12
7 prop_binSR_electron2016M_bin12
8 prop_binSR_muon2017_bin11
9 prop_binSR_muon2018_bin10
10 prop_binSR_electron2018_bin11
11 prop_binSR_muon2016M_bin11
12 prop_binSR_muon2018_bin9
13 prop_binSR_muon2017_bin10
14 prop_binSR_muon2018_bin6
15 prop_binSR_electron2018_bin10
16 prop_binSR_electron2017_bin11
17 prop_binSR_muon2018_bin8
18 prop_binSR_muon2018_bin7
19 prop_binSR_muon2016M_bin10
20 prop_binSR_muon2018_bin5
21 prop_binSR_muon2017_bin9
22 prop_binSR_muon2017_bin6
23 prop_binSR_electron2018_bin9
24 prop_binSR_electron2017_bin10
25 prop_binSR_electron2016M_bin11
26 prop_binSR_muon2017_bin8
27 prop_binSR_electron2018_bin6
28 prop_binSR_muon2018_bin4
29 lumi_2018
30 prop_binSR_muon2016M_bin9
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

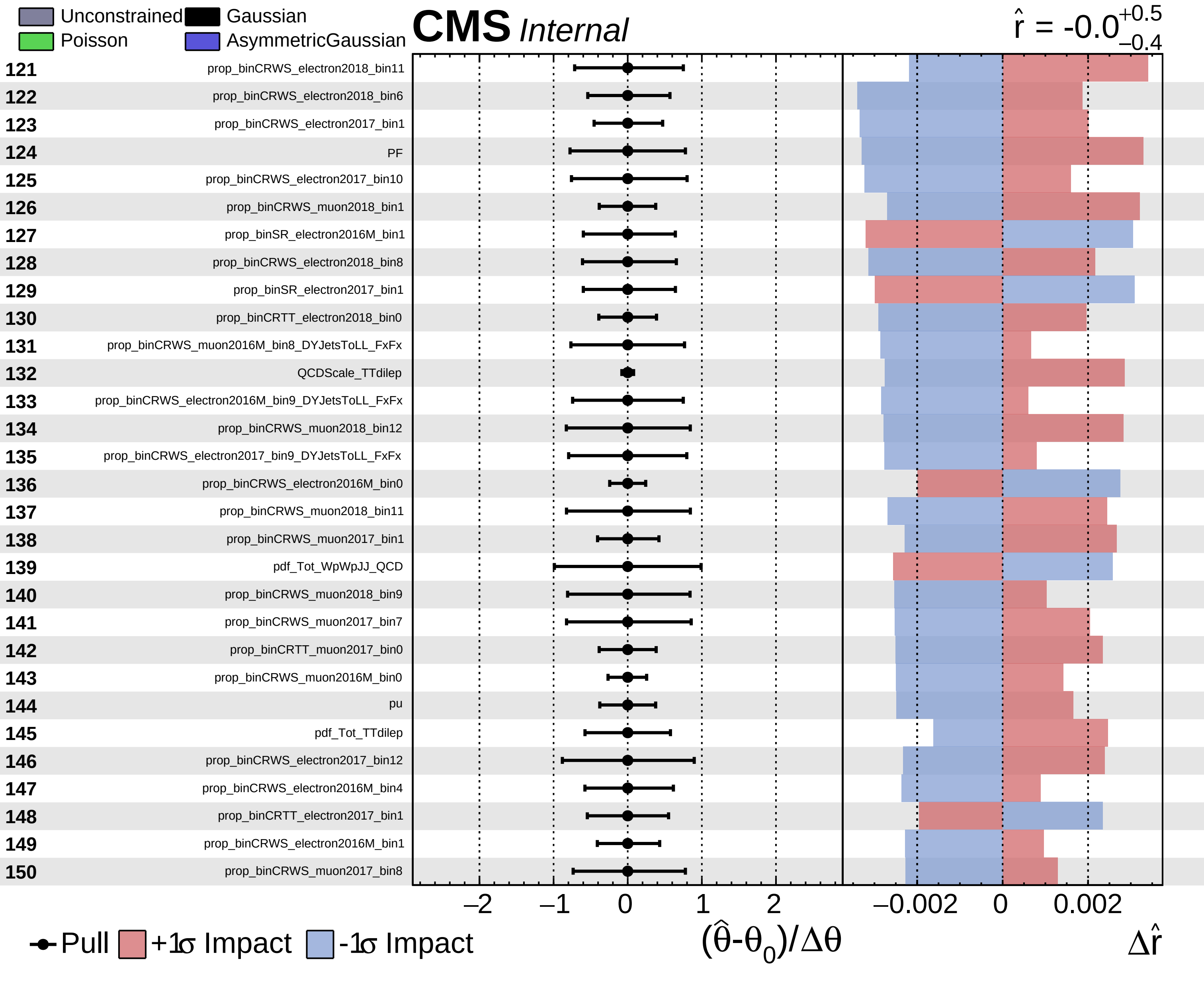


CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
31 prop_binSR_muon2016M_bin6
32 prop_binSR_electron2017_bin9
33 prop_binSR_electron2018_bin8
34 prop_binSR_electron2016M_bin10
35 prop_binSR_muon2017_bin7
36 prop_binSR_muon2018_bin3
37 prop_binSR_muon2016M_bin8
38 prop_binSR_electron2017_bin6
39 prop_binSR_electron2018_bin7
40 QCDScale_DY
41 prop_binSR_muon2017_bin4
42 prop_binSR_muon2018_bin2
43 prop_binSR_muon2017_bin5
44 prop_binSR_electron2016M_bin9
45 prop_binSR_electron2018_bin5
46 lumi_2017
47 QCDScale_WpWpJJ_QCD
48 prop_binSR_muon2016M_bin5
49 prop_binSR_electron2017_bin8
50 prop_binSR_electron2018_bin4
51 prop_binSR_muon2016M_bin7
52 prop_binSR_muon2017_bin3
53 lumi_2016M
54 prop_binSR_electron2017_bin5
55 prop_binSR_electron2016M_bin6
56 prop_binSR_muon2016M_bin4
57 jer_2018
58 prop_binSR_electron2016M_bin8
59 prop_binSR_muon2016M_bin3
60 prop_binSR_electron2017_bin7
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

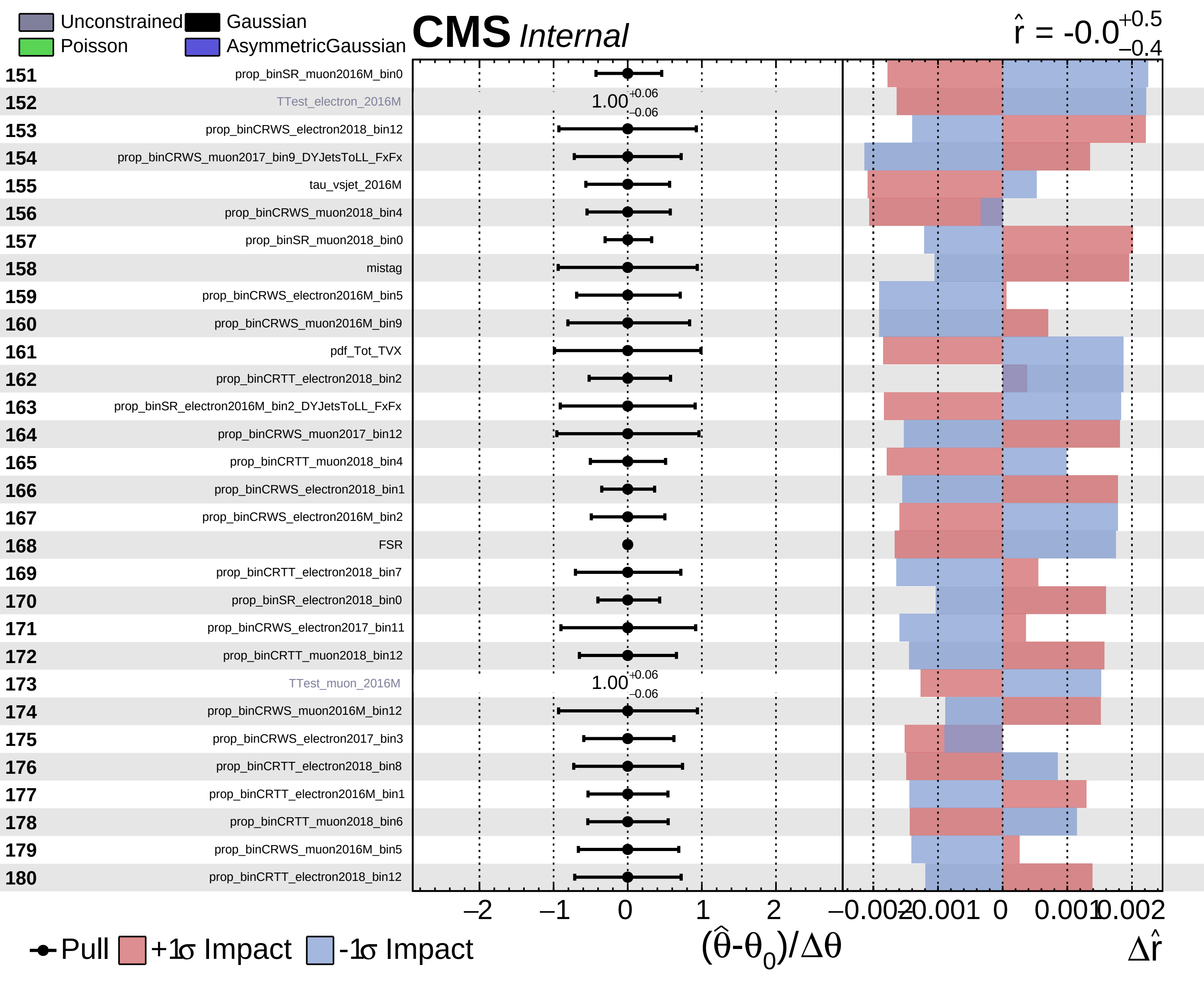

CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
61 pdf_Tot_ZZtoLep
62 prop_binCRWS_electron2018_bin0
63 prop_binSR_electron2016M_bin5
64 prop_binSR_muon2017_bin2
65 prop_binCRWS_muon2016M_bin2
66 prop_binCRWS_muon2018_bin3
67 prop_binCRWS_muon2017_bin0
68 prop_binSR_electron2016M_bin7
69 prop_binSR_electron2017_bin4
70 QCDScale_WrongSign
71 prop_binSR_muon2016M_bin2
72 prop_binSR_muon2018_bin1
73 prop_binSR_electron2018_bin3
74 prop_binSR_muon2017_bin1
75 prop_binCRWS_electron2018_bin2
76 prop_binCRWS_muon2016M_bin1
77 prop_binCRWS_electron2018_bin5
78 prop_binCRWS_electron2017_bin4
79 prop_binCRWS_electron2018_bin7
80 QCDScale_TVX
81 QCDScale_Triboson
82 prop_binCRTT_muon2018_bin0
83 prop_binSR_electron2018_bin2
84 prop_binSR_electron2016M_bin4
85 prop_binCRWS_muon2018_bin2
86 prop_binSR_electron2017_bin2
87 metUnclust
88 QCDScale_VG
89 prop_binCRWS_muon2017_bin4
90 prop_binCRWS_muon2017_bin2
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

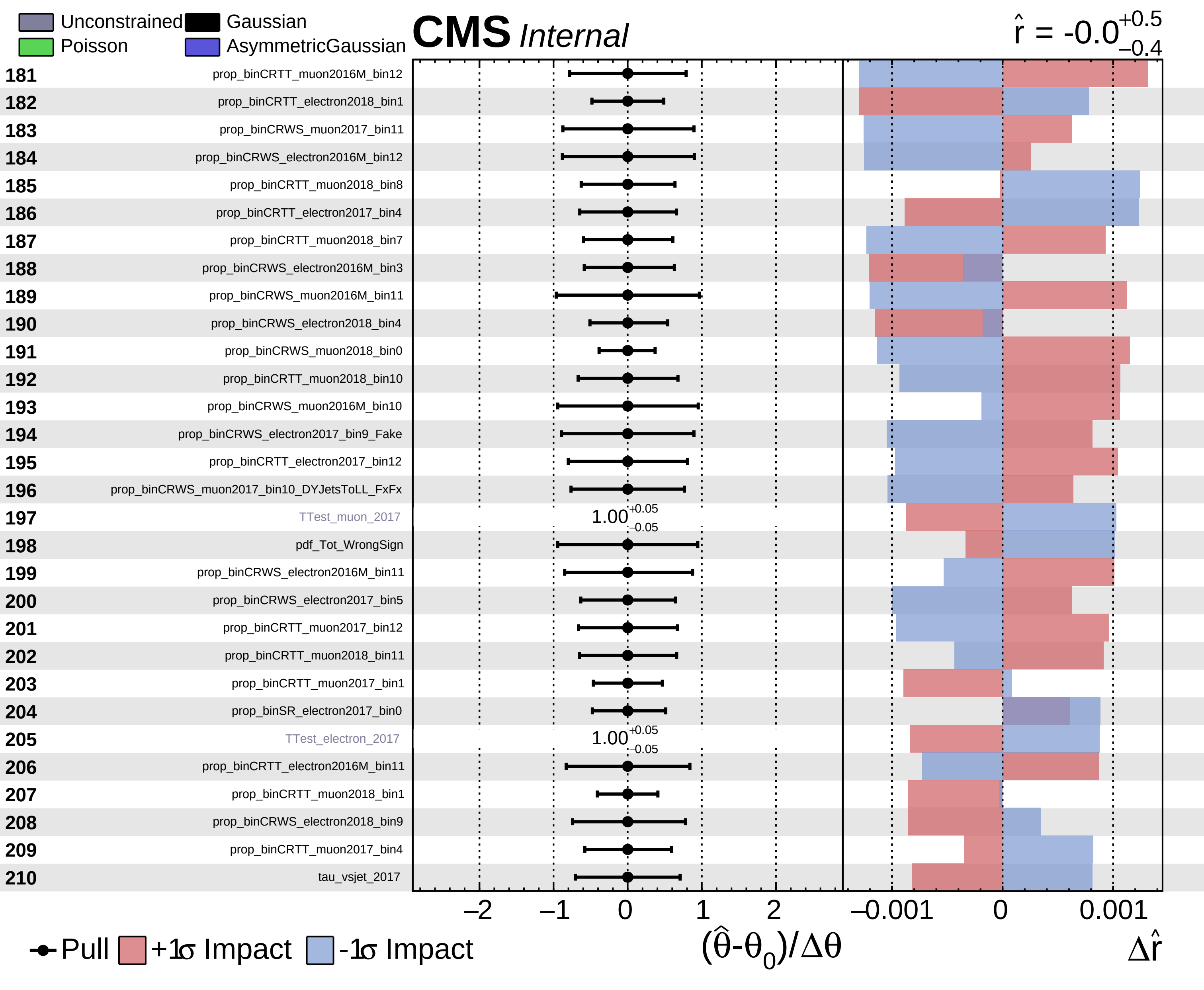


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
91 btag
92 prop_binCRWS_muon2018_bin7
93 prop_binCRWS_electron2017_bin8
94 prop_binSR_electron2017_bin3
95 prop_binCRWS_muon2017_bin3
96 prop_binCRWS_muon2018_bin8
97 prop_binCRWS_muon2018_bin6
98 prop_binCRWS_electron2017_bin0
99 pdf_Tot_DY
100 prop_binSR_muon2016M_bin1
101 jes
102 prop_binSR_electron2016M_bin2_Fake
103 prop_binCRTT_electron2017_bin0
104 prop_binCRWS_electron2017_bin2
105 prop_binCRWS_muon2017_bin6
106 prop_binSR_muon2017_bin0
107 prop_binCRWS_electron2016M_bin7_DYJetsToLL_FxFx
108 prop_binCRWS_muon2018_bin5
109 prop_binCRWS_muon2016M_bin4
110 prop_binCRWS_electron2018_bin3
111 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
112 prop_binCRWS_muon2016M_bin6
113 TES
114 prop_binCRWS_muon2017_bin5
115 prop_binCRTT_electron2016M_bin0
116 prop_binCRWS_muon2016M_bin3
117 prop_binCRWS_muon2016M_bin7
118 prop_binCRWS_electron2018_bin10
119 prop_binSR_electron2018_bin1
120 prop_binSR_electron2016M_bin3_Fake
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

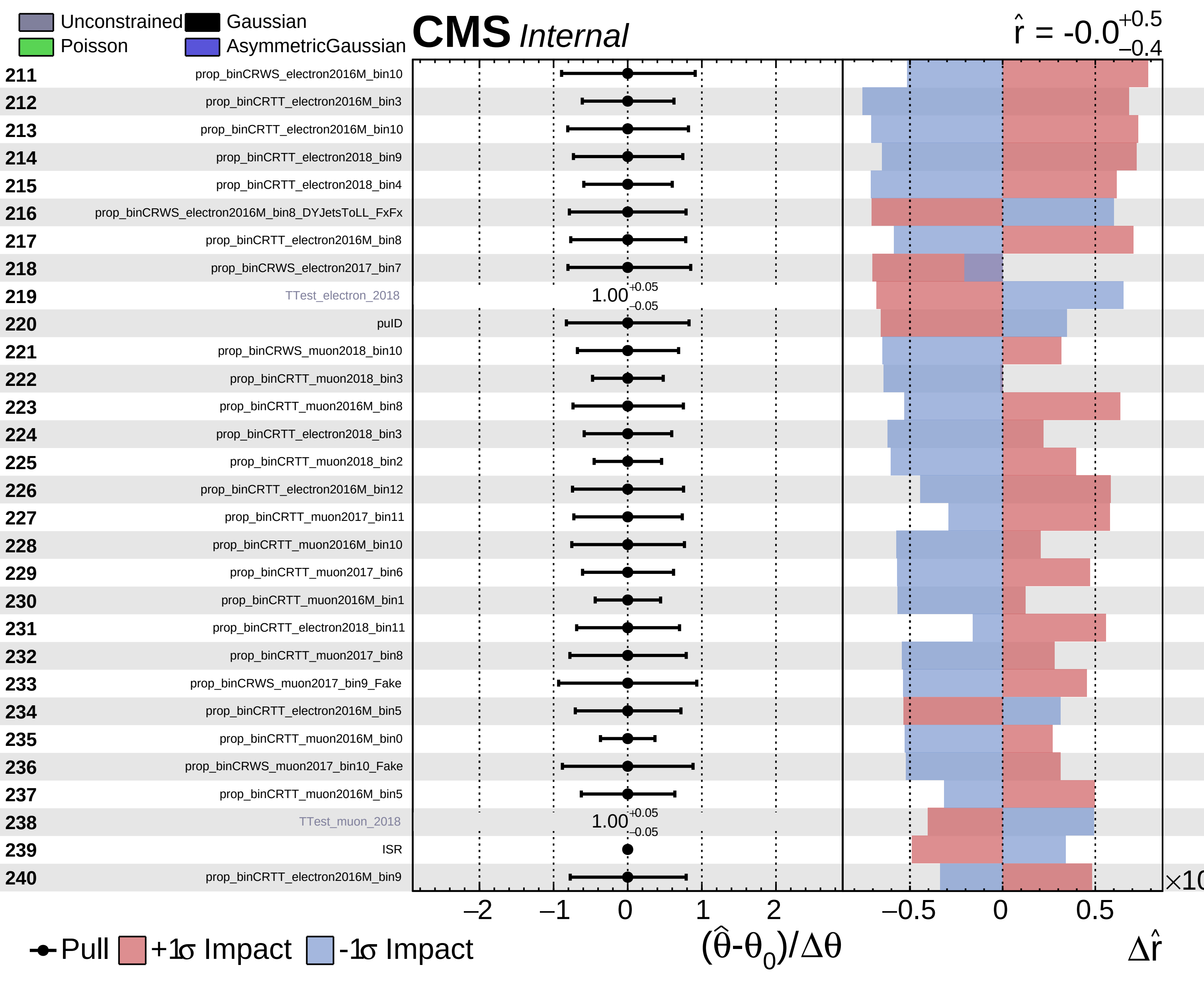


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
121 prop_binCRWS_electron2018_bin11
122 prop_binCRWS_electron2018_bin6
123 prop_binCRWS_electron2017_bin1
124 PF
125 prop_binCRWS_electron2017_bin10
126 prop_binCRWS_muon2018_bin1
127 prop_binSR_electron2016M_bin1
128 prop_binCRWS_electron2018_bin8
129 prop_binSR_electron2017_bin1
130 prop_binCRTT_electron2018_bin0
131 prop_binCRWS_muon2016M_bin8_DYJetsToLL_FxFx
132 QCDScale_TTdilep
133 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
134 prop_binCRWS_muon2018_bin12
135 prop_binCRWS_electron2017_bin9_DYJetsToLL_FxFx
136 prop_binCRWS_electron2016M_bin0
137 prop_binCRWS_muon2018_bin11
138 prop_binCRWS_muon2017_bin1
139 pdf_Tot_WpWpJJ_QCD
140 prop_binCRWS_muon2018_bin9
141 prop_binCRWS_muon2017_bin7
142 prop_binCRTT_muon2017_bin0
143 prop_binCRWS_muon2016M_bin0
144 pu
145 pdf_Tot_TTdilep
146 prop_binCRWS_electron2017_bin12
147 prop_binCRWS_electron2016M_bin4
148 prop_binCRTT_electron2017_bin1
149 prop_binCRWS_electron2016M_bin1
150 prop_binCRWS_muon2017_bin8
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

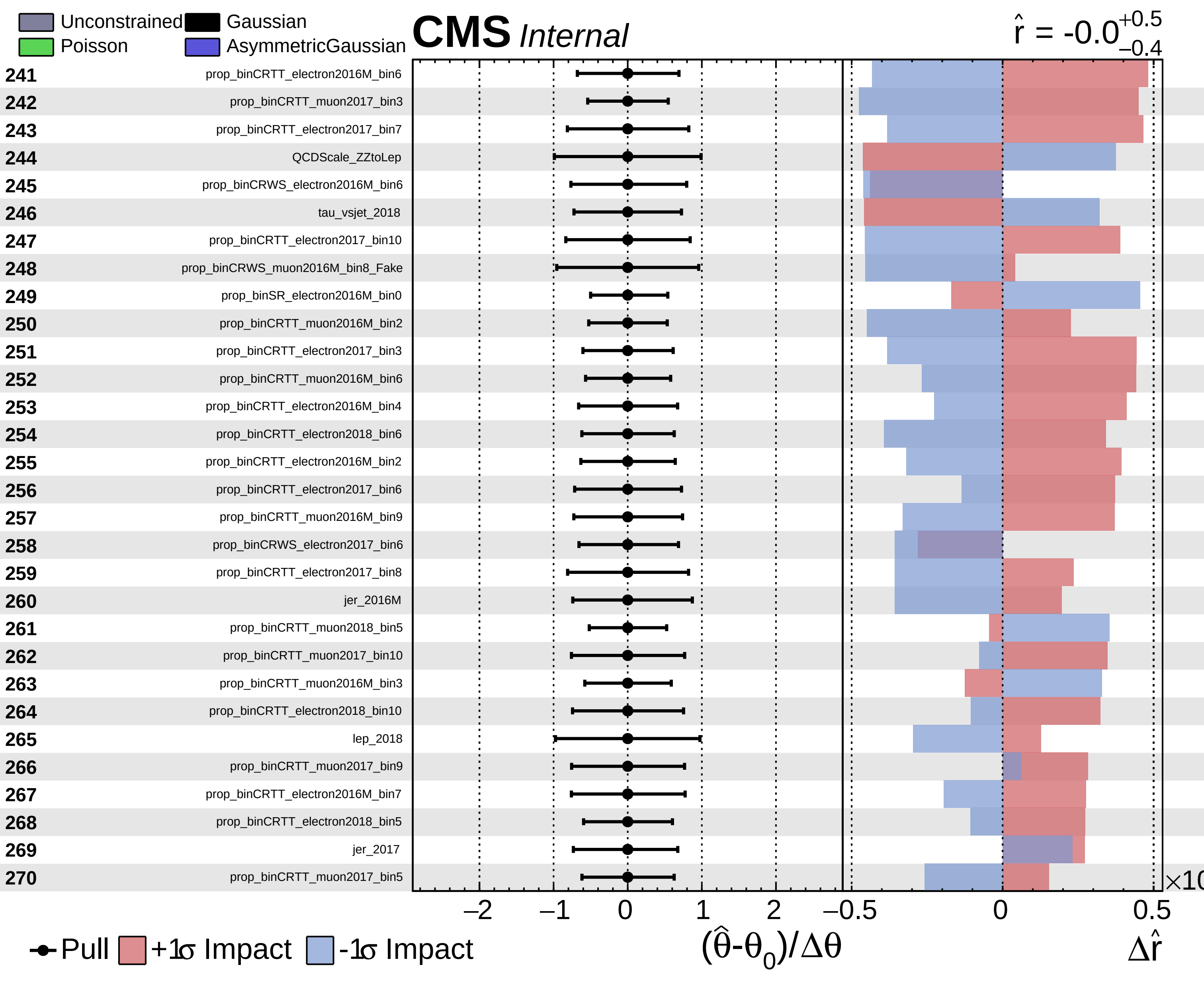

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0 +0.5 −0.4
151 prop_binSR_muon2016M_bin0
152 TTest_electron_2016M
1.00 +0.06 −0.06
153 prop_binCRWS_electron2018_bin12
154 prop_binCRWS_muon2017_bin9_DYJetsToLL_FxFx
155 tau_vsjet_2016M
156 prop_binCRWS_muon2018_bin4
157 prop_binSR_muon2018_bin0
158 mistag
159 prop_binCRWS_electron2016M_bin5
160 prop_binCRWS_muon2016M_bin9
161 pdf_Tot_TVX
162 prop_binCRTT_electron2018_bin2
163 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
164 prop_binCRWS_muon2017_bin12
165 prop_binCRTT_muon2018_bin4
166 prop_binCRWS_electron2018_bin1
167 prop_binCRWS_electron2016M_bin2
168 FSR
169 prop_binCRTT_electron2018_bin7
170 prop_binSR_electron2018_bin0
171 prop_binCRWS_electron2017_bin11
172 prop_binCRTT_muon2018_bin12
173 TTest_muon_2016M
1.00 +0.06 −0.06
174 prop_binCRWS_muon2016M_bin12
175 prop_binCRWS_electron2017_bin3
176 prop_binCRTT_electron2018_bin8
177 prop_binCRTT_electron2016M_bin1
178 prop_binCRTT_muon2018_bin6
179 prop_binCRWS_muon2016M_bin5
180 prop_binCRTT_electron2018_bin12
−2
−1
0
1
2
−0.002
−0.001
0
0.001
0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

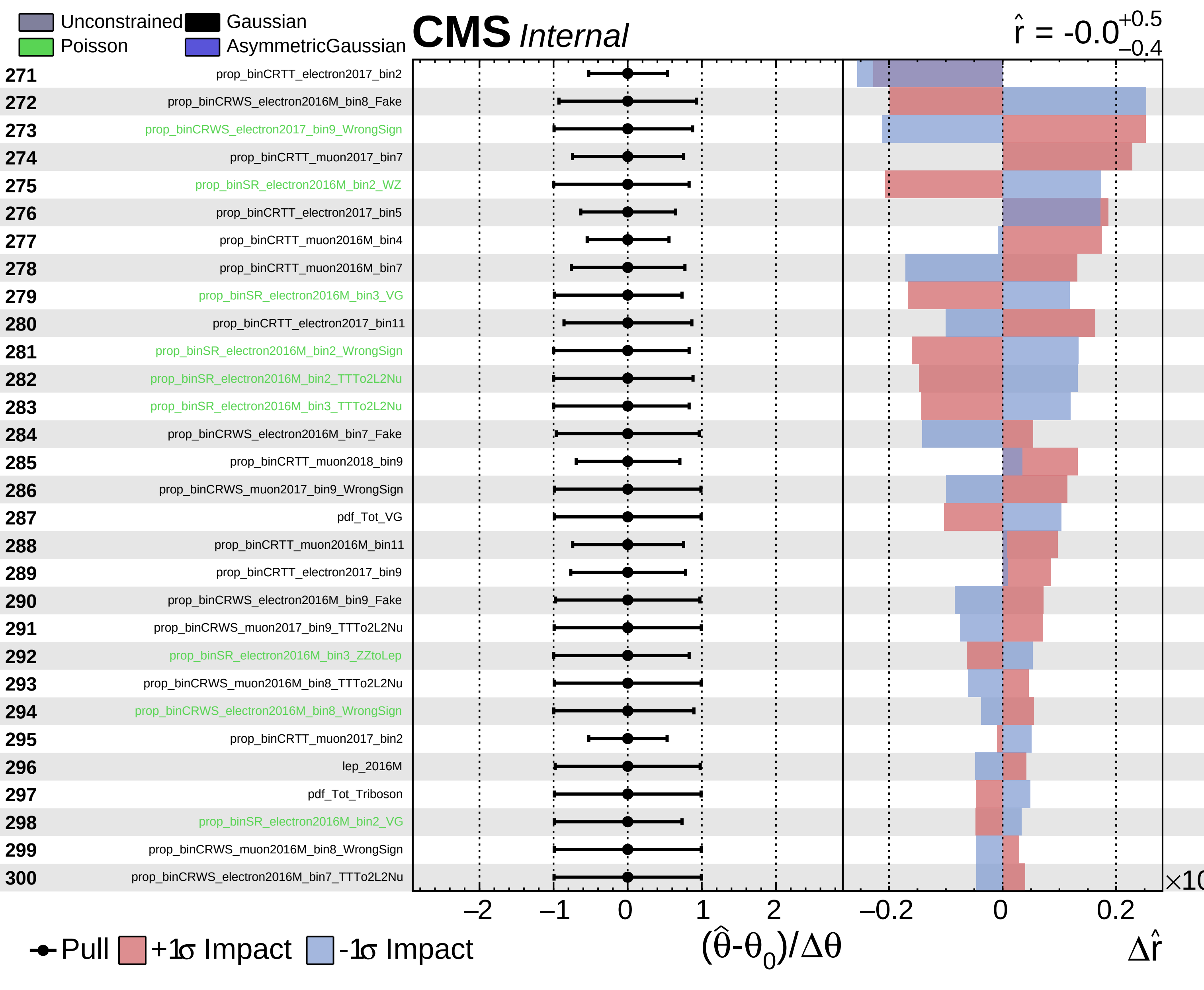


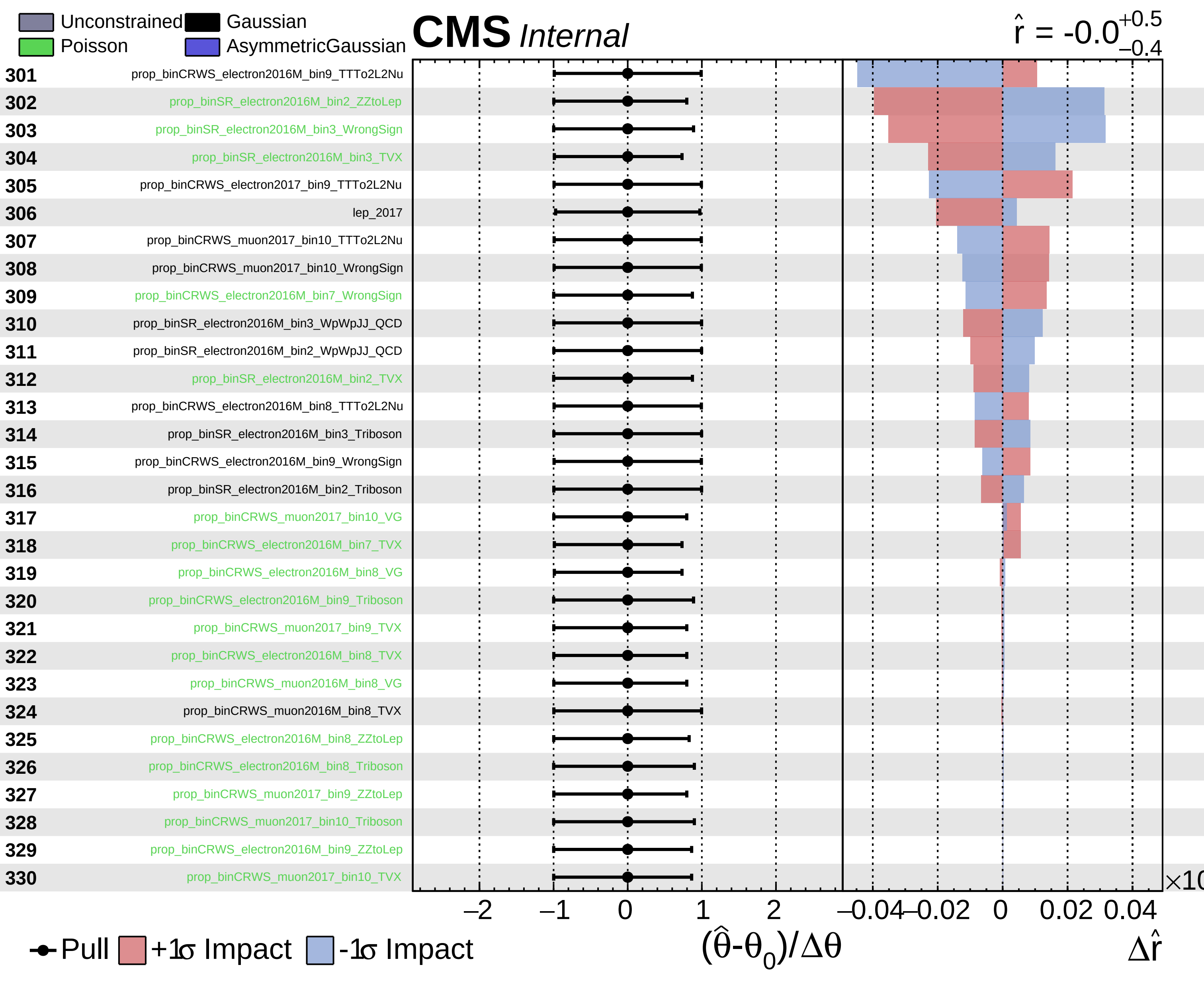
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
181 prop_binCRTT_muon2016M_bin12
182 prop_binCRTT_electron2018_bin1
183 prop_binCRWS_muon2017_bin11
184 prop_binCRWS_electron2016M_bin12
185 prop_binCRTT_muon2018_bin8
186 prop_binCRTT_electron2017_bin4
187 prop_binCRTT_muon2018_bin7
188 prop_binCRWS_electron2016M_bin3
189 prop_binCRWS_muon2016M_bin11
190 prop_binCRWS_electron2018_bin4
191 prop_binCRWS_muon2018_bin0
192 prop_binCRTT_muon2018_bin10
193 prop_binCRWS_muon2016M_bin10
194 prop_binCRWS_electron2017_bin9_Fake
195 prop_binCRTT_electron2017_bin12
196 prop_binCRWS_muon2017_bin10_DYJetsToLL_FxFx
197 TTest_muon_2017
$1.00^{+0.05}_{-0.05}$
198 pdf_Tot_WrongSign
199 prop_binCRWS_electron2016M_bin11
200 prop_binCRWS_electron2017_bin5
201 prop_binCRTT_muon2017_bin12
202 prop_binCRTT_muon2018_bin11
203 prop_binCRTT_muon2017_bin1
204 prop_binSR_electron2017_bin0
205 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
206 prop_binCRTT_electron2016M_bin11
207 prop_binCRTT_muon2018_bin1
208 prop_binCRWS_electron2018_bin9
209 prop_binCRTT_muon2017_bin4
210 tau_vsjet_2017
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



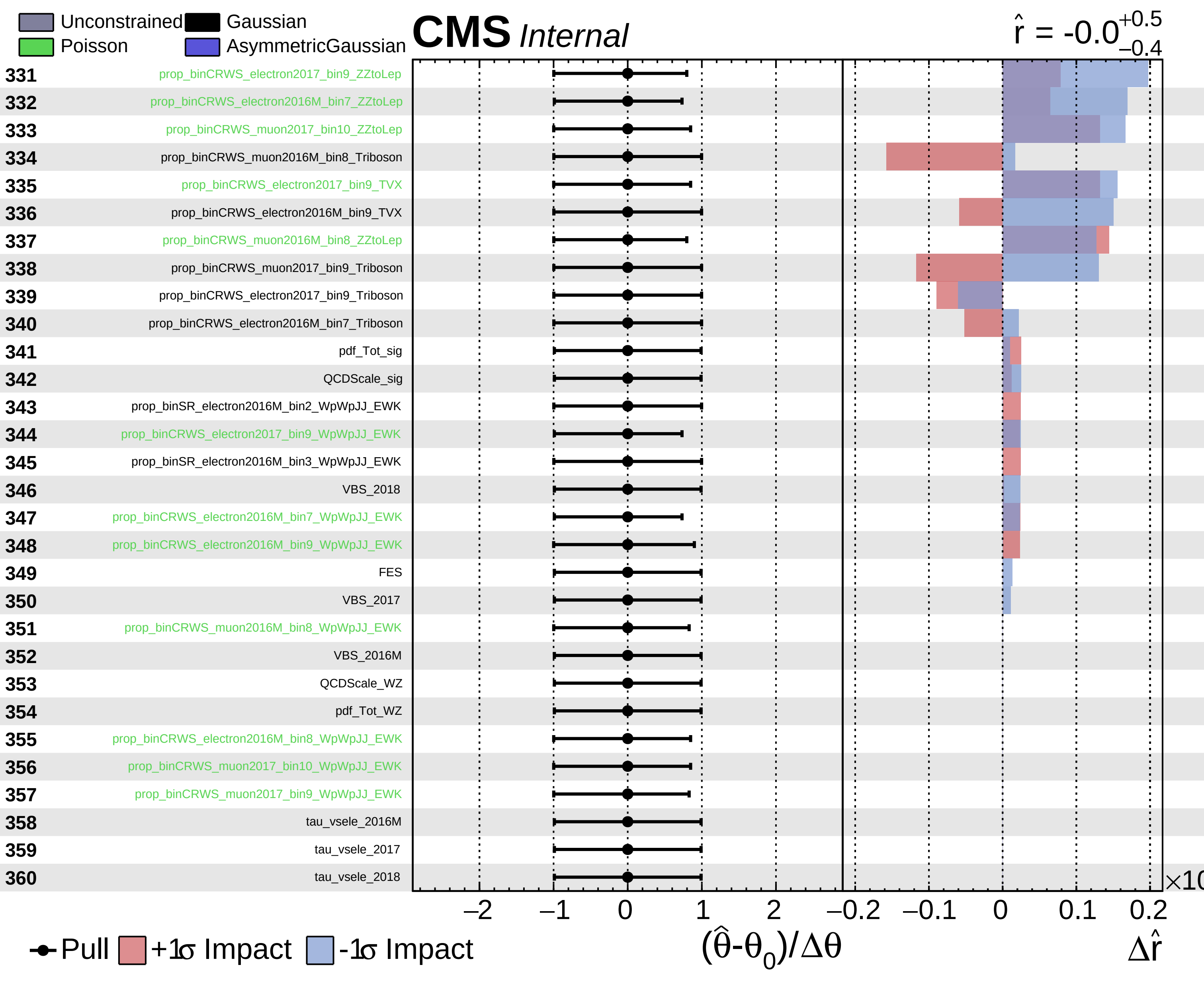
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
211 prop_binCRWS_electron2016M_bin10
212 prop_binCRTT_electron2016M_bin3
213 prop_binCRTT_electron2016M_bin10
214 prop_binCRTT_electron2018_bin9
215 prop_binCRTT_electron2018_bin4
216 prop_binCRWS_electron2016M_bin8_DYJetsToLL_FxFx
217 prop_binCRTT_electron2016M_bin8
218 prop_binCRWS_electron2017_bin7
219 TTest_electron_2018 $1.00^{+0.05}_{-0.05}$
220 puID
221 prop_binCRWS_muon2018_bin10
222 prop_binCRTT_muon2018_bin3
223 prop_binCRTT_muon2016M_bin8
224 prop_binCRTT_electron2018_bin3
225 prop_binCRTT_muon2018_bin2
226 prop_binCRTT_electron2016M_bin12
227 prop_binCRTT_muon2017_bin11
228 prop_binCRTT_muon2016M_bin10
229 prop_binCRTT_muon2017_bin6
230 prop_binCRTT_muon2016M_bin1
231 prop_binCRTT_electron2018_bin11
232 prop_binCRTT_muon2017_bin8
233 prop_binCRWS_muon2017_bin9_Fake
234 prop_binCRTT_electron2016M_bin5
235 prop_binCRTT_muon2016M_bin0
236 prop_binCRWS_muon2017_bin10_Fake
237 prop_binCRTT_muon2016M_bin5
238 TTest_muon_2018 $1.00^{+0.05}_{-0.05}$
239 ISR
240 prop_binCRTT_electron2016M_bin9
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

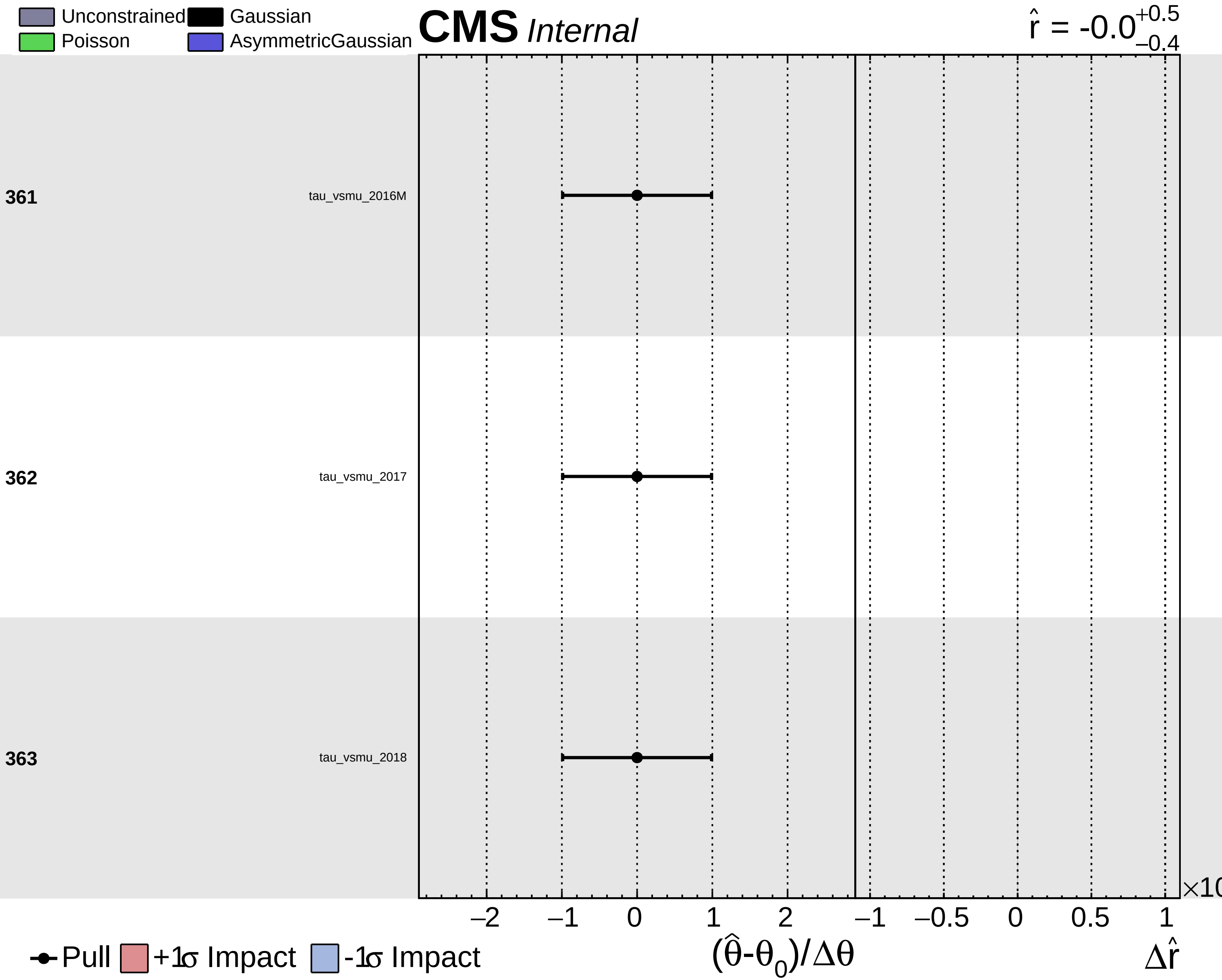

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0 +0.5 −0.4
241 prop_binCRTT_electron2016M_bin6
242 prop_binCRTT_muon2017_bin3
243 prop_binCRTT_electron2017_bin7
244 QCDScale_ZZtoLep
245 prop_binCRWS_electron2016M_bin6
246 tau_vsjet_2018
247 prop_binCRTT_electron2017_bin10
248 prop_binCRWS_muon2016M_bin8_Fake
249 prop_binSR_electron2016M_bin0
250 prop_binCRTT_muon2016M_bin2
251 prop_binCRTT_electron2017_bin3
252 prop_binCRTT_muon2016M_bin6
253 prop_binCRTT_electron2016M_bin4
254 prop_binCRTT_electron2018_bin6
255 prop_binCRTT_electron2016M_bin2
256 prop_binCRTT_electron2017_bin6
257 prop_binCRTT_muon2016M_bin9
258 prop_binCRWS_electron2017_bin6
259 prop_binCRTT_electron2017_bin8
260 jer_2016M
261 prop_binCRTT_muon2018_bin5
262 prop_binCRTT_muon2017_bin10
263 prop_binCRTT_muon2016M_bin3
264 prop_binCRTT_electron2018_bin10
265 lep_2018
266 prop_binCRTT_muon2017_bin9
267 prop_binCRTT_electron2016M_bin7
268 prop_binCRTT_electron2018_bin5
269 jer_2017
270 prop_binCRTT_muon2017_bin5
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
271 prop_binCRTT_electron2017_bin2
272 prop_binCRWS_electron2016M_bin8_Fake
273 prop_binCRWS_electron2017_bin9_WrongSign
274 prop_binCRTT_muon2017_bin7
275 prop_binSR_electron2016M_bin2_WZ
276 prop_binCRTT_electron2017_bin5
277 prop_binCRTT_muon2016M_bin4
278 prop_binCRTT_muon2016M_bin7
279 prop_binSR_electron2016M_bin3_VG
280 prop_binCRTT_electron2017_bin11
281 prop_binSR_electron2016M_bin2_WrongSign
282 prop_binSR_electron2016M_bin2_TTTo2L2Nu
283 prop_binSR_electron2016M_bin3_TTTo2L2Nu
284 prop_binCRWS_electron2016M_bin7_Fake
285 prop_binCRTT_muon2018_bin9
286 prop_binCRWS_muon2017_bin9_WrongSign
287 pdf_Tot_VG
288 prop_binCRTT_muon2016M_bin11
289 prop_binCRTT_electron2017_bin9
290 prop_binCRWS_electron2016M_bin9_Fake
291 prop_binCRWS_muon2017_bin9_TTTo2L2Nu
292 prop_binSR_electron2016M_bin3_ZZtoLep
293 prop_binCRWS_muon2016M_bin8_TTTo2L2Nu
294 prop_binCRWS_electron2016M_bin8_WrongSign
295 prop_binCRTT_muon2017_bin2
296 lep_2016M
297 pdf_Tot_Triboson
298 prop_binSR_electron2016M_bin2_VG
299 prop_binCRWS_muon2016M_bin8_WrongSign
300 prop_binCRWS_electron2016M_bin7_TTTo2L2Nu
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
301 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
302 prop_binSR_electron2016M_bin2_ZZtoLep
303 prop_binSR_electron2016M_bin3_WrongSign
304 prop_binSR_electron2016M_bin3_TVX
305 prop_binCRWS_electron2017_bin9_TTTo2L2Nu
306 lep_2017
307 prop_binCRWS_muon2017_bin10_TTTo2L2Nu
308 prop_binCRWS_muon2017_bin10_WrongSign
309 prop_binCRWS_electron2016M_bin7_WrongSign
310 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
311 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
312 prop_binSR_electron2016M_bin2_TVX
313 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
314 prop_binSR_electron2016M_bin3_Triboson
315 prop_binCRWS_electron2016M_bin9_WrongSign
316 prop_binSR_electron2016M_bin2_Triboson
317 prop_binCRWS_muon2017_bin10_VG
318 prop_binCRWS_electron2016M_bin7_TVX
319 prop_binCRWS_electron2016M_bin8_VG
320 prop_binCRWS_electron2016M_bin9_Triboson
321 prop_binCRWS_muon2017_bin9_TVX
322 prop_binCRWS_electron2016M_bin8_TVX
323 prop_binCRWS_muon2016M_bin8_VG
324 prop_binCRWS_muon2016M_bin8_TVX
325 prop_binCRWS_electron2016M_bin8_ZZtoLep
326 prop_binCRWS_electron2016M_bin8_Triboson
327 prop_binCRWS_muon2017_bin9_ZZtoLep
328 prop_binCRWS_muon2017_bin10_Triboson
329 prop_binCRWS_electron2016M_bin9_ZZtoLep
330 prop_binCRWS_muon2017_bin10_TVX
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
331 prop_binCRWS_electron2017_bin9_ZZtoLep
332 prop_binCRWS_electron2016M_bin7_ZZtoLep
333 prop_binCRWS_muon2017_bin10_ZZtoLep
334 prop_binCRWS_muon2016M_bin8_Triboson
335 prop_binCRWS_electron2017_bin9_TVX
336 prop_binCRWS_electron2016M_bin9_TVX
337 prop_binCRWS_muon2016M_bin8_ZZtoLep
338 prop_binCRWS_muon2017_bin9_Triboson
339 prop_binCRWS_electron2017_bin9_Triboson
340 prop_binCRWS_electron2016M_bin7_Triboson
341 pdf_Tot_sig
342 QCDScale_sig
343 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
344 prop_binCRWS_electron2017_bin9_WpWpJJ_EWK
345 prop_binSR_electron2016M_bin3_WpWpJJ_EWK
346 VBS_2018
347 prop_binCRWS_electron2016M_bin7_WpWpJJ_EWK
348 prop_binCRWS_electron2016M_bin9_WpWpJJ_EWK
349 FES
350 VBS_2017
351 prop_binCRWS_muon2016M_bin8_WpWpJJ_EWK
352 VBS_2016M
353 QCDScale_WZ
354 pdf_Tot_WZ
355 prop_binCRWS_electron2016M_bin8_WpWpJJ_EWK
356 prop_binCRWS_muon2017_bin10_WpWpJJ_EWK
357 prop_binCRWS_muon2017_bin9_WpWpJJ_EWK
358 tau_vsele_2016M
359 tau_vsele_2017
360 tau_vsele_2018
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
361
tau_vsmu_2016M
362
tau_vsmu_2017
363
tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$